

Phase 7: Adaptive Behaviour

Introduction and objective of phase 7

Phase 7 introduces adaptive behaviour through feedback collection, storage, and response adaptation.

What Was Missing in Phase 6

Phase 6 ignored user feedback and could not adapt future responses based on user satisfaction.

Before Phase 7: The agent always behaved the same way.

After Phase 7: The agent changes its response style based on user feedback. So Phase 7 extends your Phase 6 Banking AI Support & Advisory Agent by introducing:

Feedback → Storage → Adaptation → Behaviour Change

1. Added Feedback Storage

```
FEEDBACK_FILE = "feedback.json"
```

```
feedback_data = {  
    "helpful": 0,  
    "not_helpful": 0  
}
```

Stores user feedback across sessions.

2. New Functions added

Load Feedback Function (load_feedback())

The system loads feedback from persistent storage so that adaptation continues across sessions.

Save Feedback Function (save_feedback())

The updated feedback counters are stored in feedback.json for future interactions.

Record Feedback Function (record_feedback(feedback))

The agent records whether users marked responses as helpful or not helpful.

3. Adaptation Logic Added

Adaptive Logic

If negative feedback reaches three or more instances, the response style changes from NORMAL to DETAILED.

Before Behaviour

Identity proof, address proof, income proof.

After Behaviour

Home loan documents generally include identity proof, address proof, income proof, bank statements, and property documents.

4. Added Feedback Commands

We can now type feedbacks such as the following:

feedback helpful

feedback not helpful

5. Demo of the feedback

Customer: What documents are required for a home loan?

Agent:

What loan amount would you like to borrow?

Customer: 20 Lakh

Agent:

For a ₹20 lakh home loan, the current interest rate is **8.25% per annum**.

Approx. annual interest: **₹1,65,000**.

Customer: feedback not useful

Agent:

Thank you for your feedback. Future responses may adapt based on this feedback.

Customer: feedback not useful

Agent:

Thank you for your feedback. Future responses may adapt based on this feedback.

Customer: feedback not useful

Agent:

Thank you for your feedback. Future responses may adapt based on this feedback.

Customer: What documents are required for a home loan?

Agent:

What loan tenure would you prefer?

Customer: 20 years

Agent:

Could you please clarify what you mean by "20 years"?

6. New Failure Modes added such as

1. Feedback abuse
2. Global adaptation affecting all users
3. Rule-based adaptation instead of true learning

7. Conclusion

Phase 7 introduced adaptive behaviour into the Banking AI Support & Advisory Agent by enabling the system to collect, store, and use feedback signals. Feedback is persisted in a feedback.json file and influences future response generation. When repeated negative feedback exceeds a threshold, the agent automatically changes from concise responses to more detailed explanatory responses. This demonstrates behaviour adaptation based on previous user interactions.